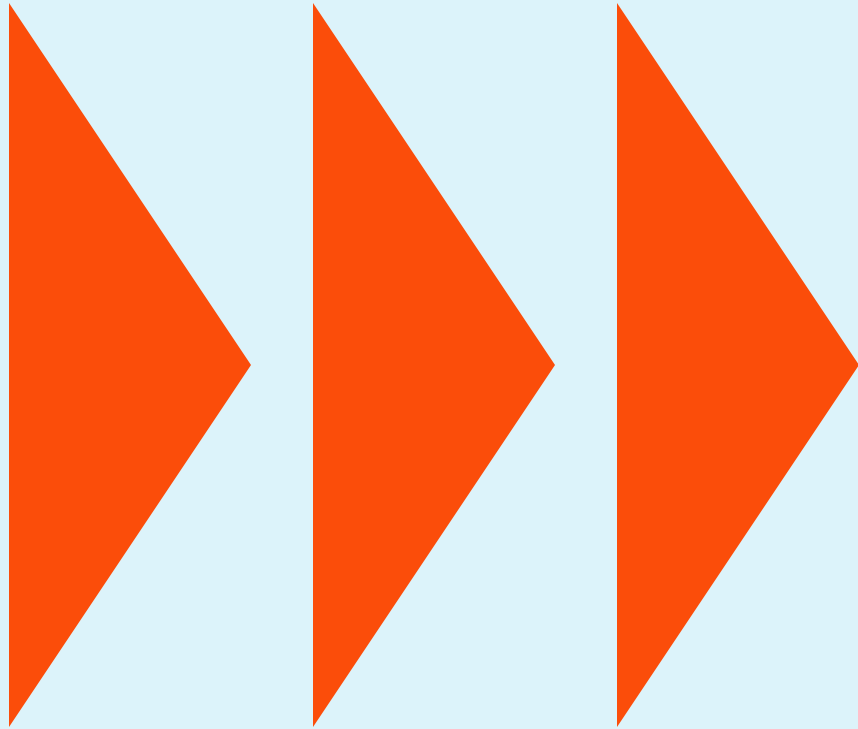




EXL Profile



Summary

AI/ML & Generative AI engineer with **3+ years** hands-on experience building production-grade **LLM systems**, **RAG pipelines**, and **agentic workflows** for insurance, analytics and enterprise use cases. Designed end-to-end solutions including full-stack RAG claim verification, Text-to-SQL with **self-correction**, and ML pipelines across regression, anomaly detection, and computer vision. Proficient in Python, SQL, FastAPI, Docker, **LangChain**, **LangGraph**, vector databases and modern LLMs (Gemini/Groq). Strong engineering foundation from **Oracle** across automation, CI/CD pipelines, and reliability improvements across Linux/Windows environments.

Full-Stack GenAI System for Insurance Claim Verification

- **Objective:** Automated end-to-end health insurance claim evaluation by validating uploaded medical bills against **5 policy PDFs** and **7 exclusion categories** using a **multi-agent, policy-grounded RAG** pipeline. Generated explainable approve/deny verdicts with controlled LLM reasoning and reduced manual claim-review dependency through agentic orchestration.
- **Data Source and Data Preparation:** Ingested **5 policy PDFs** and uploaded bills/claim inputs by chunking policy text, generating **HuggingFace Sentence Transformers** dense embeddings and **BM25 sparse vectors (pinecone-text)** for **Pinecone** hybrid indexing, structuring bill signals for disease/expense extraction + exclusion mapping, and creating a small offline test-claims set for retrieval/response validation.
- **Model Building and Prediction:** Implemented a retrieval + reasoning pipeline combining semantic retrieval with **Gemini LLM** control to extract diseases/expenses, validate exclusions, and generate evidence-backed decisions. Enhanced retrieval with **hybrid search (dense + BM25)** and a **cross-encoder re-ranker** to improve chunk relevance. Rebuilt the adjudication flow as a **LangGraph multi-agent** system with **Extraction, Policy, and Adjudication** agents orchestrated in a controlled graph for stepwise decisioning.
- **Deployment:** Shipped a production-ready full-stack app with **FastAPI + Uvicorn** (backend) and **React + Vite + Tailwind** (frontend), runnable via **Docker/Docker Compose**; integrated **LangSmith** tracing across agent nodes, LLM calls, and retrieval steps (latency/token visibility) and optimized a multi-stage Docker build to reduce image size from **919MB → 455MB (~50%)**.
- **Business Impact:** Improved claim evaluation using policy-grounded reasoning with **hybrid retrieval (dense + sparse/BM25) + cross-encoder re-ranking**, **LangGraph** orchestration with graceful failure handling, and **RAGAS** checks (Context Precision, Answer Relevancy) to reduce hallucinations while cutting Docker image size by **~50%**.

Natural Language to SQL AI System with Self-Correction

- **Objective:** Built a production-ready Natural Language → SQL system that lets non-technical users query databases safely by generating executable SQL, enforcing **SQL guardrails**, automatically **self-correcting** failed queries, and returning database-backed answers through a FastAPI + Streamlit application.
- **Data Source and Data Preparation:** Used a **MySQL 8.0** database with pre-seeded schema and source data (initialized via SQL + CSV assets) and incorporated schema context into the generation pipeline to ensure column/table awareness. Added safety validation rules to detect and block unsafe/undesired SQL patterns before execution, constraining runtime behavior to prevent destructive or malicious queries.
- **Model Building and Prediction:** Implemented a LangChain-based SQL generation chain using **Gemini LLM**, followed by a **SQL Guardrail** stage and a **Self-Correction Loop** that re-prompts and repairs SQL when execution fails. Executed generated SQL against the real MySQL database to produce grounded answers and included an optional **agentic workflow** using **LangGraph (ReAct-style)** with SQL tools to inspect schema, reason step-by-step, execute queries iteratively, and expose intermediate tool calls for transparency.
- **Deployment:** Delivered an interactive interface using **Streamlit** with a **FastAPI (Uvicorn)** backend and containerized the complete stack using **Docker** (UI + API + database) for reproducible local and cloud deployment. Applied a multi-stage Docker strategy with CPU-only optimization and layered build cleanup, reducing the final image size from **2.1GB → 516MB (~75%)** while keeping the database bootstrapped and ready-to-run.
- **Business Impact:** Increased Text-to-SQL trust and reliability via **real DB execution**, **SQL safety guardrails**, and an **auto-repair loop**, validated with **RAGAS** (faithfulness, answer relevance, Context Precision/Recall) achieving **0.86+ faithfulness**, and improved operational efficiency with **~75%** smaller deployment artifacts.

Other Projects

- **Lead Conversion Propensity Modeling (Regression):** Built and mechanized a **Scikit-learn regression** model on **1.2M+ records** to predict lead conversion propensity with **91% accuracy**, enabling automated lead scoring and improving marketing decision-making.
- **Solar Panel Defect Classification (Computer Vision):** Developed an **EfficientNet**-based classifier for **6 defect classes** on **800+** solar panel images and deployed a **Streamlit** app for real-time inference with confidence scoring, improving accuracy from **56% → 82%** using transfer learning and **17+ hours** of hyperparameter tuning.
- **Real-Time Anomaly Detection for Server Downtime:** Deployed a real-time anomaly detection pipeline using **Oracle AI** and **LSTM/RNN** models to predict US-region server downtime from load metrics and benchmarked it against traditional ML baselines achieving **~11% faster detection**.
- **Stock Movement Prediction:** Built an **LSTM time-series** model using historical market data to predict stock movement, achieving **92%+ test accuracy** and ranking **1st among 50+ teams** in a predictive modeling competition.



Education & Technical Skills

Key Skills: Python, SQL, C++, Machine Learning, Deep Learning, Computer Vision, RAG, LLMs, GenAI, Agentic Architectures (**LangChain**, **LangGraph**), Prompt Engineering, Hugging Face, Scikit-learn, TensorFlow/Keras, PyTorch, XGBoost, FastAPI, MySQL, Linux, System Design, Docker, Git, Streamlit, CI/CD pipelines, Pinecone, SentenceTransformers

Education: B.Tech — Electrical Engineering, Indian Institute of Technology Bombay, India

Certifications: Data Structures & Algorithms, Data Science & AI Masters – From Python to GenAI from Udemy

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